

Cobalt



What is it?

Cobalt is a silver, gray, magnetic, brittle metal. It is used to produce blue colors in pottery, glass, porcelain and enamel. The most common source of exposure is nickel-plated objects, which usually contain cobalt. Cobalt is used as a binding agent to make hard metals, such as carbides. These are in items such as rock drills, cutting tools, drawing, pressing, and stamping tools. Some hard metals have been reported to contain as much as 10% cobalt. A reaction to cobalt often occurs at the same time as a nickel allergy.

COBALT can also be known as:

- Cobalt Blue
- Cobalt (II) Chloride Hexahydrate
- Cobalt Dichloride
- Cobaltous Chloride

Some Possible Uses of COBALT/Where it might be found:

- Snaps, zippers, buttons, buckles
- Leather
- Animal feeds
- Antiperspirants
- Blue porcelain
- Bricklaying
- Carbide industry
- Cement
- Paint, resins
- Enamels
- Dental appliances
- Foam stabilizer in beer
- Electroplating
- Vitamin B12 supplement
- Fertilizer and feed additive

- Tools, utensils, instruments
- Hair dyes
- Cosmetics
- Hand etching and grinding
- Invisible ink
- Jewelry
- Joint replacements
- Machinery parts
- Paint industry
- Pigment in some artistry paints
- Pigments in pottery, glass, and crayons
- Polyester resin manufacture
- Polyester resins
- Pottery workers
- Printing inks
- Shell splinters
- Tattoo ink. Used less commonly in modern tattoos.

General sources:

- Nickel plated objects
- Some hair dyes (light brown)
- Blue dyes, pigments, enamels, oil colors
- Printing ink: drying agent
- Linseed oil: drying agent, vitamin B12, prostheses and implants

Industrial sources:

- Carbides: cutting tools, drills
- Alloys: alnico, duralumin, elgiloy, nobelium, permalloy, stellite, ticonium, vitallium
- Cement, bricks: mostly in Europe
- Paint industry: blue pigments, drier for lacquers, varnishes, enamels
- Polyester resins: cobalt naphthenate as drier and catalyst

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- Pottery workers: contact with wet clay, porcelain dyes
- Electroplating: to brighten nickel plated objects
- Electrical industry: lamp filaments, magnets
- Animal feed: used as mineral supplement
- Textiles: as mordant
- Automotive: additive in lubricating oils

Other products that might react like COBALT:

- Amine cobalt salt
- Cyanocobalamin

What should I do to avoid COBALT?

The best thing a cobalt allergic person can do is to remove all cobalt containing items from use. If cobalt containing items must be used, a covering of some type should be placed over the metal. Masking tape or clear varnish are good covering materials. Remember that these coverings will wear off, usually in 2-3 months of use. Keep in mind that cobalt may go through rubber gloves, with or without cotton liners.

Cobalt and nickel are often found in the same or similar products. Items that contain cobalt may have a darker metal color.

As many sources of cobalt do not have an ingredient list, it is necessary to test metal items for cobalt release. This can be done by swabbing with a test kit that contains 2-nitroso-1-naphthol-4-sulfonic acid. If the liquid on the swab turns yellow-orange or peach in color, cobalt is present.

A cobalt test kit (Reveal and Conceal Cobalt spot test kit) can be obtained from SmartPractice (www.smartpractice.com) or Amazon (amazon.com). Another source is Dormer (dormer.com; Cobalt spot test). Testing an item for cobalt does not harm the item and can be done in the store prior to purchase.

There are many sources for hypoallergenic jewelry and earrings. Consult your jeweler or search for jewelry on your favorite search engine online. One source is www.simplywhispers.com.

Reference: *Dermatitis* 2016;27:3-8.

In addition, the ACDS Contact Allergen Management Program® (CAMP®) can help manage contact dermatitis. It helps to find products that are safe to use by creating a list of products that do not contain one's allergens. For more information about CAMP®, please ask your doctor or visit www.contactderm.org.

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